

Row Echelon form

Row-Echelon Form (Short Notes)

- Using row operations, a matrix can be converted into **row-echelon form**.
- A matrix may contain **zero rows** (rows with all elements = 0).
- A non-zero row has at least one **non-zero element**.

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

ROW ECHELON FORM

Definition:

A matrix is said to be in **Row Echelon Form** if it satisfies the following conditions:

1. Zero Rows at Bottom

All rows containing only zeros must be placed **below** the non-zero rows.

Example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Leading Element (Pivot) Rule

The first non-zero element in each row is called a **pivot**.

All elements **below the pivot** must be zero.

3. Staircase Pattern

The pivot of each successive row must be placed **to the right** of the pivot in the previous row, forming a **step/staircase shape**.

Example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

Simple Idea:

- ✓ Zero rows at bottom
- ✓ Zeros below each pivot
- ✓ Pivots move right like stairs

Conclusion:

Thus, Row Echelon Form arranges a matrix in a simplified staircase form, making it easier to solve linear equations.

Example 1.10

Reduce the matrix $\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$ to a row-echelon form.

Solution

$$\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix} \xrightarrow[\text{Operations Panel}]{\begin{matrix} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{matrix}} \begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow[\text{Operations Panel}]{R_3 \rightarrow R_3 - \frac{1}{2}R_2} \begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{Solution} \\ \text{(Row-Echelon Form)} \end{matrix}$$

NOTE

$$\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix} \xrightarrow[\text{Alternative Row-Echelon Form}]{R_2 \rightarrow R_2/8} \begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

A further simplification gives another form. This confirms that the row-row-echelon form is not unique.

So, a row-echelon form of a matrix is **not necessarily unique**.



Example 1.14

Reduce the matrix $\begin{bmatrix} 0 & 3 & 1 & 6 \\ -1 & 0 & 2 & 5 \\ 4 & 2 & 0 & 0 \end{bmatrix}$ to a row-echelon form.

SOLUTION: STEP-BY-STEP REDUCTION

$$\begin{aligned}
 &\begin{bmatrix} 0 & 3 & 1 & 6 \\ -1 & 0 & 2 & 5 \\ 4 & 2 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} -1 & 0 & 2 & 5 \\ 0 & 3 & 1 & 6 \\ 4 & 2 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 4R_1} \begin{bmatrix} -1 & 0 & 2 & 5 \\ 0 & 3 & 1 & 6 \\ 0 & 2 & 8 & 20 \end{bmatrix} \\
 &\xrightarrow{R_3 \rightarrow R_3 - \frac{2}{3}R_2} \begin{bmatrix} -1 & 0 & 2 & 5 \\ 0 & 3 & 1 & 6 \\ 0 & 0 & \frac{22}{3} & 16 \end{bmatrix} \xrightarrow{R_3 \rightarrow 3R_3} \begin{bmatrix} -1 & 0 & 2 & 5 \\ 0 & 3 & 1 & 6 \\ 0 & 0 & 22 & 48 \end{bmatrix}.
 \end{aligned}$$

THE FINAL MATRIX IS IN ROW-ECHELON FORM

Ex 1.2

$$\left[\sum_m^{\sigma} \right]$$

Question 1:

Find the rank of the following matrices by the minor method:

(i) $\begin{bmatrix} 0 & 0 & 3 & 1 & 6 \\ -1 & -1 & 0 & 2 & 5 \\ 0 & 4 & 2 & 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$

Solution:

(i) $A = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ A is of order 3×4 . $\Rightarrow \rho(A) \leq 3$

Now $|A| = \begin{vmatrix} 2 & -4 \\ -1 & 2 \end{vmatrix} = 0 \Rightarrow \rho(A) \neq 2.$

So $\rho(A) = 1$

Now in A a minor of order 1 is $a_{11} = 2 \neq 0$

So $\rho(A) = 1$

$$\begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 3 \\ 8 & 1 & 2 \end{vmatrix} = 8(3-2) = 8(1) = 8 \neq 0$$

$$A = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -4 \\ -1 & 2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 3 \\ 8 & 1 & 2 \end{vmatrix}$$

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -7 \end{bmatrix}$$

(ii) Let

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -7 \end{bmatrix}$$

(ii) A is of order 3×2 . So $\rho(A) \leq \text{minimum of } 3 \times 2$. (i.e) $\rho(A) \leq 2$.

Consider a minor of order $3 \times 3 = \begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = 7 - 12 = -5 \neq 0$
So $\rho(A) = 2$.

(ii) A is order 3×2 .

$$3 = \begin{vmatrix} -2 & 3 \\ -1 & 3 \end{vmatrix} \begin{vmatrix} 1 & 3 \\ 4 & -7 \end{vmatrix}$$

THANK YOU !